

BIVEK PRADHAN

*Assistant Professor of
Physics | Ph.D. Scholar*

5+
YEARS TEACHING

5
PUBLICATIONS

5
CONFERENCE TALKS

CONTACT

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SKILLS

Physics Solar & plasma
physics, MHD theory
(Advanced)

Computation C, Python,
MATLAB, Mathematica,
PLUTO, MHD Code

Documentation LaTeX, MS
Word, MS PowerPoint
(Advanced)

Professional
Communication, mentoring,
academic leadership

LANGUAGES

Nepali — Native
English — Fluent
Hindi — Fluent

RESEARCH INTERESTS

- Coronal heating & MHD turbulence
- Physics-informed neural networks
- Non-linear dynamics
- AI/ML in physical modelling
- Undergraduate research mentorship

PROFESSIONAL SUMMARY

Assistant Professor of Physics and Ph.D. scholar specializing in MHD wave dynamics, compressible plasma turbulence, and solar coronal heating. Author of five peer-reviewed publications, including four first-author papers in Solar Physics and the Journal of the Korean Astronomical Society, and presenter at five international/national conferences. Combines five years of undergraduate teaching with continuous pedagogical development through UGC- and NIEPA-affiliated FDPs. Skilled in numerical modelling (PLUTO, Python, MATLAB, Mathematica) and extending this work into physics-informed neural networks (PINNs) for coronal seismology.

EDUCATION

Ph.D. in Physics (Pursuing) 2021 – Present

Sikkim Manipal Institute of Technology, Sikkim Manipal University, Sikkim, India
Thesis: “To Study the Compressibility of Magnetohydrodynamic (MHD) Waves in Perspective of Turbulence Heating of Solar Corona.”

M.Sc. Physics 2018 – 2020

Sikkim Manipal Institute of Technology, Sikkim Manipal University, Sikkim, India
Specialization: Plasma and High Energy Physics

- Dissertation (2021): “Heat Transfer through Plasma Sheath due to Multi-ion and Electron Fluxes in the Presence of Secondary Electron Emission” — modelled how a charged sheath layer near electron-emitting walls reduces heat flux from plasma to the wall.

B.Sc. Physics 2015 – 2018

Sikkim Government College, Tadong, Sikkim University, Sikkim, India

ACADEMIC & RESEARCH EXPERIENCE

Assistant Professor of Physics & NEP Coordinator 2021 – Present

Sikkim Government Science College, Chakung, Sikkim, India

- Deliver undergraduate courses across classical mechanics, electromagnetism, and modern/plasma physics
- Mentor students on academic projects and early-stage research, including computational and numerical methods
- Coordinate NEP 2020 implementation for the department, including curriculum alignment and MOOC/SWAYAM course blueprinting
- Complete UGC- and NIEPA-affiliated FDPs on pedagogy, blended learning design, and generative AI in education

M.Sc. Dissertation

Sikkim Manipal Institute of Technology, India

- Independent research project on heat transfer through a plasma sheath due to multi-ion and electron fluxes with secondary electron emission — modelled how the sheath layer suppresses heat flux to electron-emitting walls.

Summer Research Intern (SSP-2019) 2019

Institute for Plasma Research (IPR), Gandhinagar, Gujarat, India

- Independent research project, “Electromagnetic Effect on Temperature Gradient Driven Unstable Waves,” under IPR’s Summer Students Programme

PUBLICATIONS

Parametric Decay Instability as a Source of Turbulence in an Open Leakage Permitting Vertical Solar Coronal Domain

Solar Physics — doi.org/10.1007/s11207-026-02662-w

2026

Study of parametric decay instability as a turbulence-generating mechanism in an open, leakage-permitting vertical solar coronal domain.

The Dynamics of Compressible Acoustic Waves and Torsional Alfvén Waves in Perspective of Solar Coronal Heating

Journal of the Korean Astronomical Society

2026

Numerical investigation of wave propagation and energy transport by compressible acoustic and torsional Alfvén waves in an idealized 1-D solar flux tube.

A Critical Review on Turbulent Energy Dissipation Mechanism in Solar Corona to Astrophysical Systems and a Physics-Informed Neural Network Approach

Solar Physics — doi.org/10.1007/s11207-025-02583-0

2025

Reviews turbulent energy dissipation in MHD systems relevant to coronal heating and evaluates physics-informed neural networks as an emerging modelling approach.

Alfvén Wave Mode Coupling as a Source of Compressible Acoustic Waves in the Solar Corona

CRC Press (Applied Research in Engineering Sciences)

2025

Investigates the generation of compressible acoustic waves through Alfvén wave mode coupling in the solar corona.

A Linear Analysis of Torsional Alfvén Waves in Open Twisted Divergent Magnetic Flux Tubes for Coronal Heating

Solar Physics — doi.org/10.1007/s11207-024-02372-1

2024

Linear analysis of torsional Alfvén wave behaviour and energy transfer in an open, twisted, divergent flux tube, showing intermediate wave frequencies most efficiently heat the solar corona.

CONFERENCE PRESENTATIONS

“The Dynamics of Compressible Acoustic Waves and Torsional Alfvén Waves in Perspective of Solar Coronal Heating” (Oral)

International Conference on Futuristic Aspects in Science & Engineering (ICFAISE-2025), ICFAI University, Jaipur

6–7 Feb 2025

“Alfvén Wave Mode Coupling as a Source of Compressible Acoustic Waves in the Solar Corona” (Oral)

2nd International Conference on Applied Research in Engineering Sciences (ICARES-2025), Ramaiah Institute of Technology, Bengaluru

20–21 Feb 2025

“Effect of Plasma Sheath with Secondary Electron Emission on the Ti–N Chemical Bond Formation in Titanium Dental Implantation” (Oral)

1st International Conference on Physics of Materials and Technology (ICPMT-2022), Dept. of Physics, SMIT, Sikkim Manipal University (Virtual)

16–18 Feb 2022

“Computational Optimization of Plasma Parameters for Plasma Surface” (Oral)

Second International Conference on Advances in Plasma Science and Technology (ICAPST-21)

27–29 May 2021

8th PSSI Plasma Scholars Colloquium (PSC 2020) (Oral)

Plasma Science Society of India (PSSI) with KIIT Deemed to be University, Bhubaneswar, Odisha (Online)

8–9 Oct 2020

PROFESSIONAL DEVELOPMENT (FDPS & WORKSHOPS)

NEP 2020 Orientation & Sensitization Programme — Malaviya Mission Teacher Training Programme (MMTTP)

UGC-Malaviya Mission Teacher Training Centre, NIEPA, New Delhi, with Pandit Deendayal Upadhyaya Adarsha Mahavidyalaya, Sribhumi, Assam

19–30 Mar 2026

AI Pragyan: Faculty Development Programme on Generative AI — technical & pedagogical competencies for educators

SmartBridge in collaboration with IBM SkillsBuild

16 Feb – 6 Mar 2026

Design, Develop and Deliver Online/Blended Learning Courses

National Institute of Educational Planning and Administration (NIEPA) in collaboration with Sikkim University

12–16 Dec 2022

State Level Workshop on Implementation of NEP 2020 & its Preparedness

IQAC and Dept. of Education, Sikkim Government College, Burtuk, with Directorate of Higher Education (Govt. of Sikkim) and Dept. of Education, Sikkim University

10 May 2024

Mentoring for Institutional Social Responsibility and Facilitation for Community Engagement

Mahatma Gandhi National Council of Rural Education, MoE, Govt. of India, Hyderabad — sponsored by the Malaviya Mission on Teachers and Teaching

8–12 Feb 2022